



COMPUTER SECURITY

AES encryption

BY

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<https://github.com/223719/Computer-Security-Project-.git>



Problem Statement

- Normal communication systems can be attacked or monitored.
- This project demonstrates secure communication and cyberattack simulation.





Encrypted text messages

MITM attack simulation

Voice streaming

Tampering detection

INTRODUCTION



MATHEMATICAL MODEL

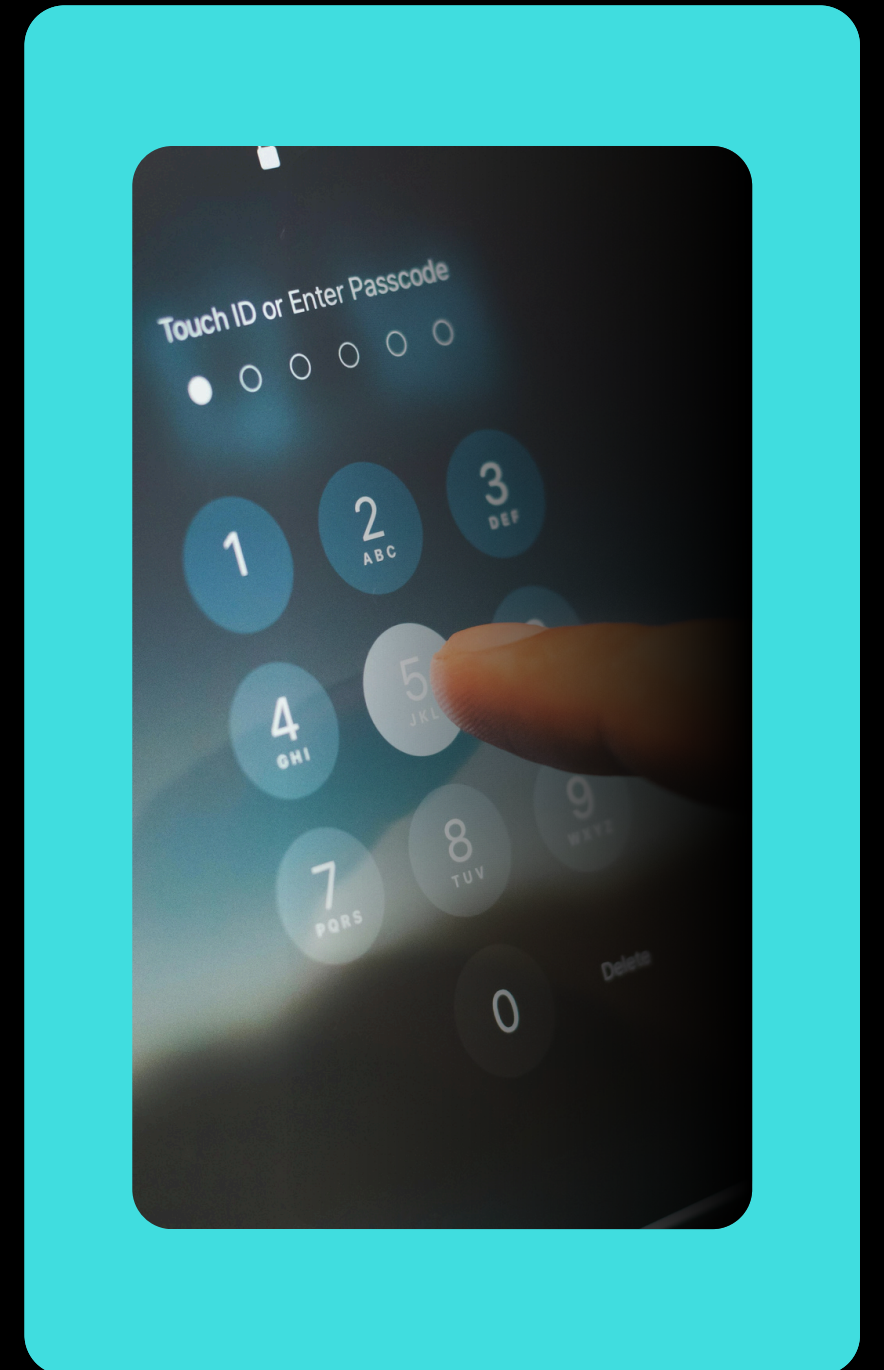
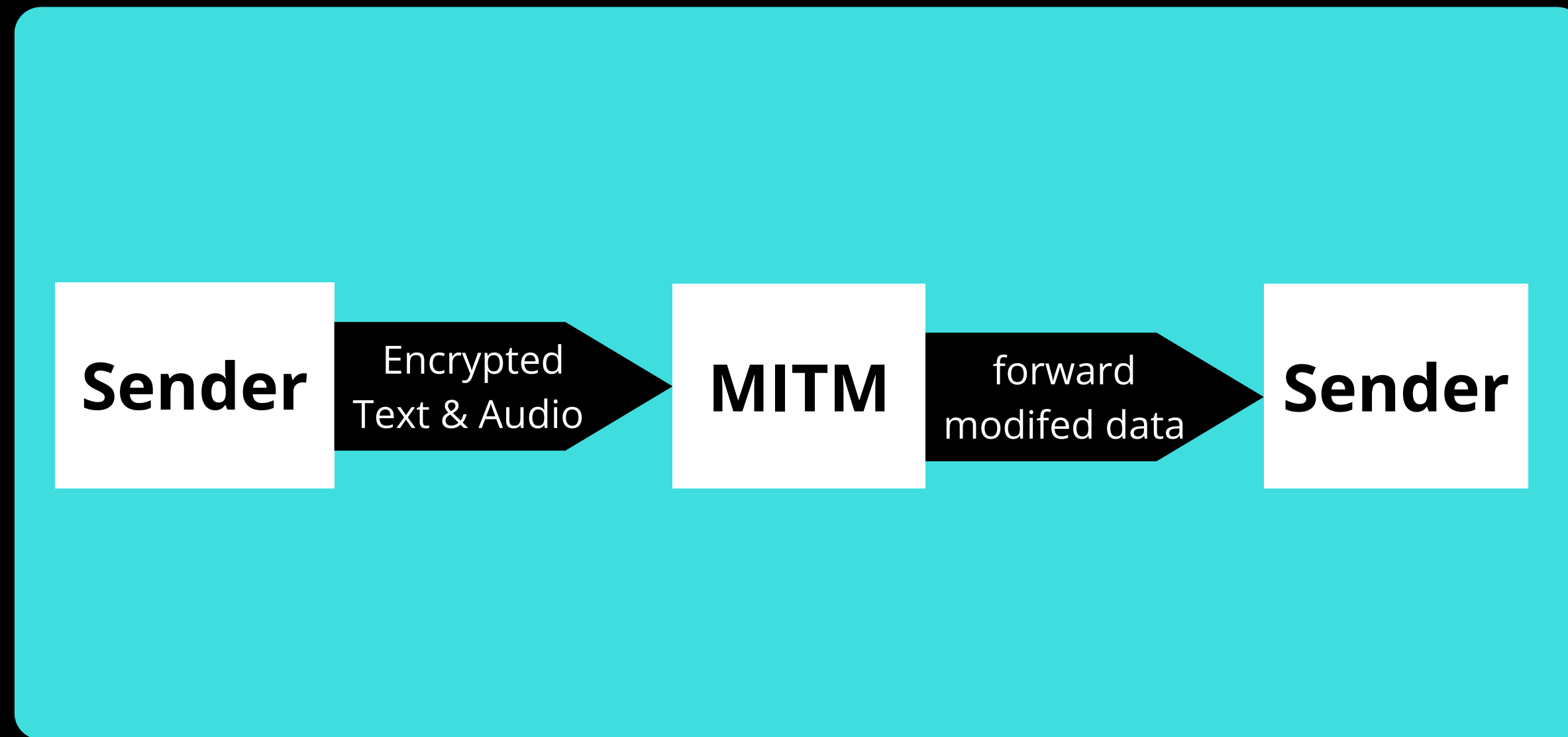
Algorithms Used

- *AES-128 Encryption*
- *SHA-256 Hashing*
- *TCP Socket Communication*





ARCHITECTURE DIAGRAM





RESULTS

- **Secure encrypted communication**
- **Detection of modified messages**
- **Successful MITM simulation**
- **Real-time voice transfer**

Conclusion

The project successfully combines networking, cryptography, and cybersecurity concepts in one secure communication system.



THANK YOU